

- 3 Female mosquitoes feed on human blood. Some species of mosquitoes are vectors of human pathogens. For example, female mosquitoes of the species *Aedes aegypti* transmit the pathogen that causes the disease yellow fever.

Mosquitoes have sensory receptors that can detect chemicals in the air. Mosquito repellents contain chemicals that are sprayed onto the skin of humans to prevent mosquitoes taking blood meals.

- (a) Some scientists investigated the effectiveness of different mosquito repellents using a human volunteer. The scientists carried out the investigation on this person in a laboratory.

Fig. 3.1 is a diagram of the experiment before the start of the investigation.

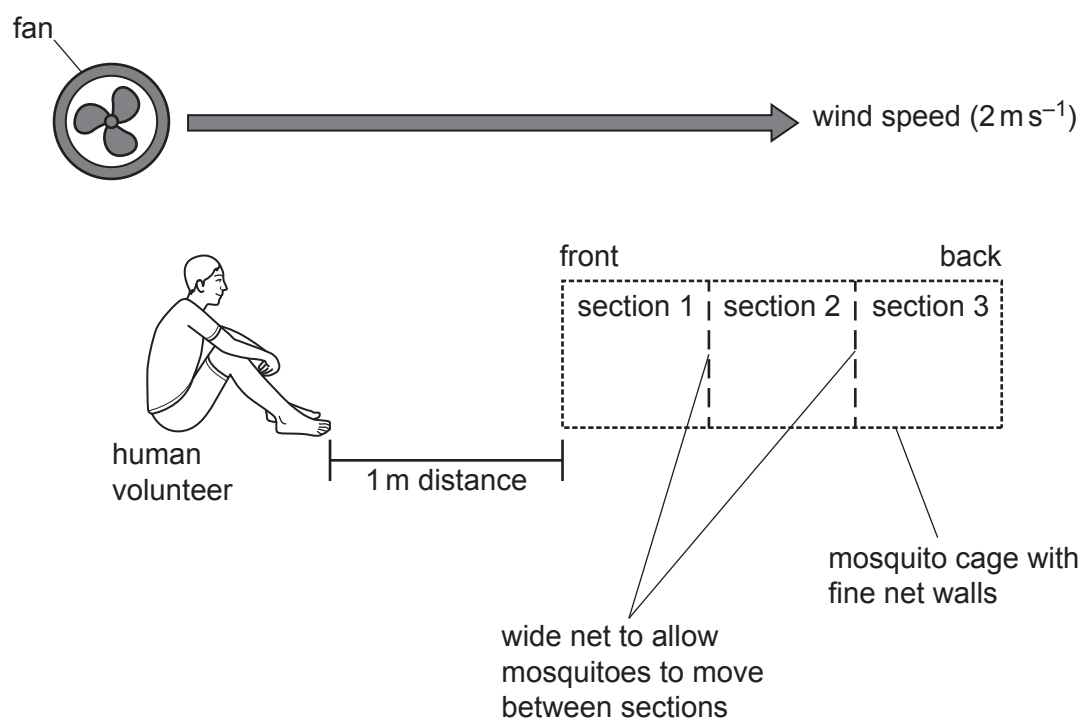


Fig. 3.1

The same procedure was used for each mosquito repellent studied.

- A human volunteer sat 1 m from a mosquito cage, as shown in Fig. 3.1.
- The person sprayed some mosquito repellent onto the skin of both arms.
- A fan was turned on so that air moved from the person towards the mosquito cage. The moving air carried chemicals, including the mosquito repellent, from the person into the mosquito cage.
- The scientists added 100 female mosquitoes of *A. aegypti* to section 2 of the mosquito cage.
- The mosquitoes were left in the mosquito cage for 15 minutes. The walls of the mosquito cage were made of fine net to prevent mosquitoes leaving the cage. The mosquitoes moved freely between sections 1, 2 and 3.
- After 15 minutes, the scientists counted the number of mosquitoes in each section of the mosquito cage. The percentage of mosquitoes in section 1 of the mosquito cage was calculated.
- This procedure was then repeated three times on different days, using the same person.

(i) State a suitable control for this investigation.

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..... [1]

(ii) Suggest **one** risk to the **scientists** when carrying out this investigation **and** state a suitable precaution they should take.

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..... [1]

(b) Table 3.1 shows the results of the investigation.

Table 3.1

chemical in mosquito repellent	percentage concentration of chemical in mosquito repellent	percentage of mosquitoes in section 1 of the mosquito cage mean $\pm$ standard error (SE)
DEET	40	68.55 $\pm$ 6.42
DEET	98	33.70 $\pm$ 4.06
lemon eucalyptus oil	30	29.62 $\pm$ 6.31
picaridin	10	78.65 $\pm$ 6.00

The results of this investigation were published in a scientific paper.

A student who read this paper concluded that the most effective mosquito repellents contained DEET or lemon eucalyptus oil.

The student was planning to travel to a country where malaria is present. The student decided to use a mosquito repellent that contained DEET to prevent malaria infection.

Explain whether the procedure **and** the data in Table 3.1 support or do **not** support this decision to use a mosquito repellent that contained DEET.

[4]

[Total: 6]

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